

Alcohol, Amine, Ether, and Thiol

UNIT XII

Learning Outcomes

At the end of each topic, the students should be able to:

- ❑ Identify and classify organic compounds, including hydrocarbons, alcohols, amines, ethers, and thiols, based on their structures and characteristic functional groups.
- ❑ Name and write the structural formulas of hydrocarbons, alcohols, amines, ethers, and thiols using appropriate IUPAC nomenclature rules.

Learning Outcomes

At the end of each topic, the students should be able to:

- ❑ Explain the physical properties of hydrocarbons, alcohols, amines, ethers, and thiols, including boiling points, melting points, polarity, and solubility.
- ❑ Describe the characteristic chemical reactions of hydrocarbons, alcohols, amines, ethers, and thiols based on their functional groups.

Alcohols

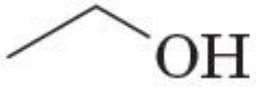
Alcohol: A compound that containing an **–OH (hydroxyl group)** bonded to a tetrahedral carbon.

- Methanol, CH_3OH, is the simplest alcohol.

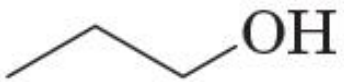
Nomenclature

1. Select the longest carbon chain that contains the –OH group as the parent alkane and number it from the end that gives the -OH the lower number.
2. Change the ending of the parent alkane from ***e*** to ***ol*** and use a number to show the location of the –OH group; for cyclic alcohols, the carbon bearing the –OH group is carbon-1.
3. Name and number substituents and list them in alphabetical order.

Nomenclature



IUPAC: Ethanol
Common: (Ethyl alcohol)



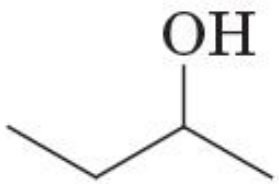
1-Propanol
(Propyl alcohol)



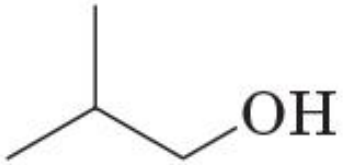
2-Propanol
(Isopropyl alcohol)



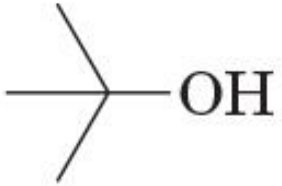
1-Butanol
(Butyl alcohol)



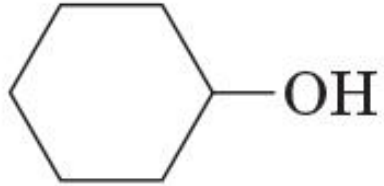
2-Butanol
(*sec*-Butyl alcohol)



2-Methyl-1-propanol
(Isobutyl alcohol)



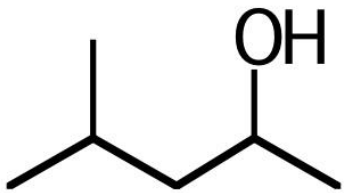
2-Methyl-2-propanol
(*tert*-Butyl alcohol)

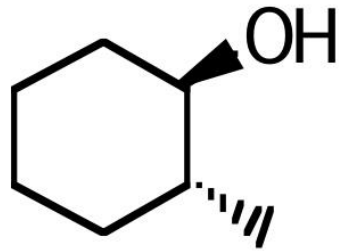


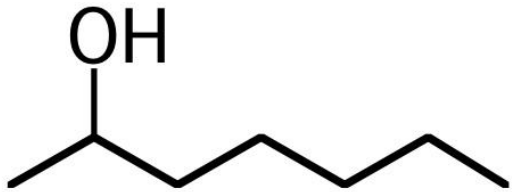
Cyclohexanol
(Cyclohexyl alcohol)

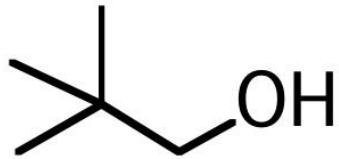
EXERCISE

Write the IUPAC name for each alcohol.

(a) 

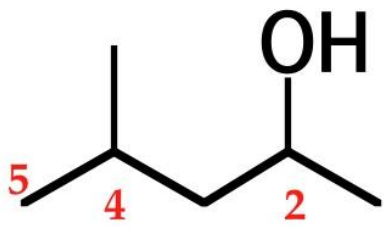
(b) 

(c) 

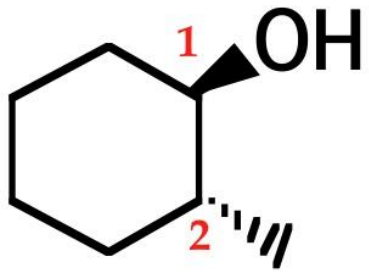
(d) 

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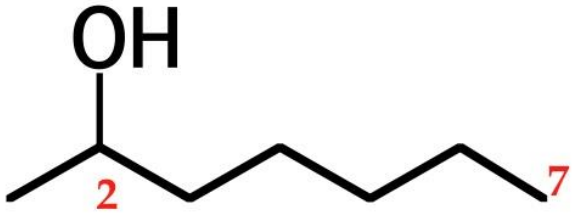
Solution:

(a) 

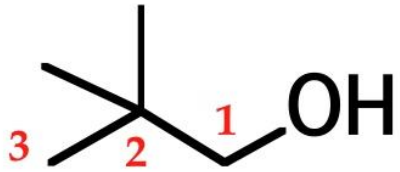
4-Methyl-2-pentanol

(b) 

trans-2-Methylcyclohexanol

(c) 

2-Heptanol

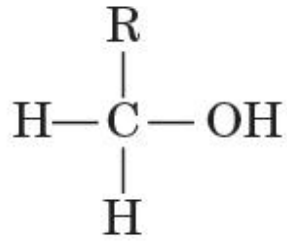
(d) 

2,2-Dimethyl-1-propanol

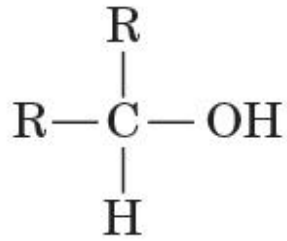
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Classification of alcohols

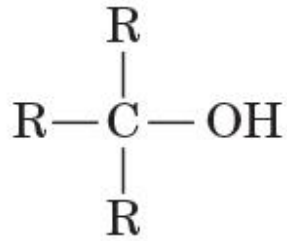
- Alcohols are classified as primary (1°), secondary (2°), or tertiary (3°), depending on the number of carbon groups bonded to the carbon bearing the –OH group.



primary (1°) alcohol
one C, shown with R



secondary (2°) alcohol
two C's, shown with two R's

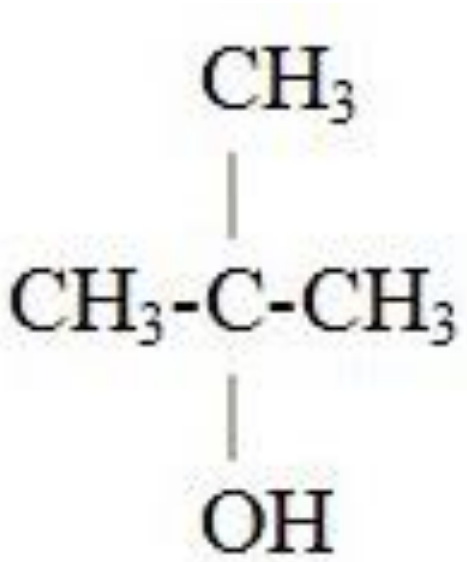


tertiary (3°) alcohol
three C's, shown with three R's

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EXERCISE

The following alcohol is classified as a _____ alcohol.



a. Primary

b. Secondary

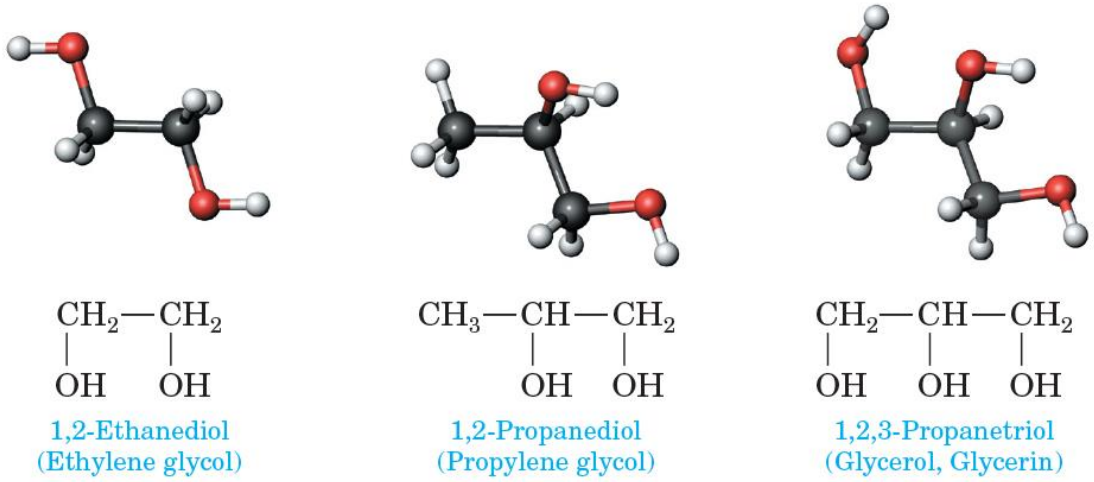
c. Tertiary

d. Quaternary

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Nomenclature

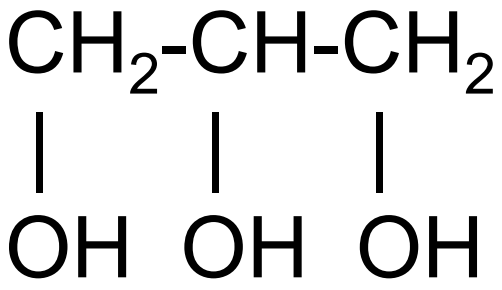
- In the IUPAC system, a compound containing two hydroxyl groups is named as a **diol,** one containing three hydroxyl groups as a **triol,** and so forth.
- IUPAC names for diols, triols, and so on retain the final "**-e**" in the name of the parent alkane.
- We commonly refer to compounds containing two hydroxyl groups on adjacent carbons as **glycols.**



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EXERCISE

What is the IUPAC name of the following alcohol?



a. 1,2,3-hydroxypropanol

b. 1,2,3-propanol

c. 1,2,3-propanetriol

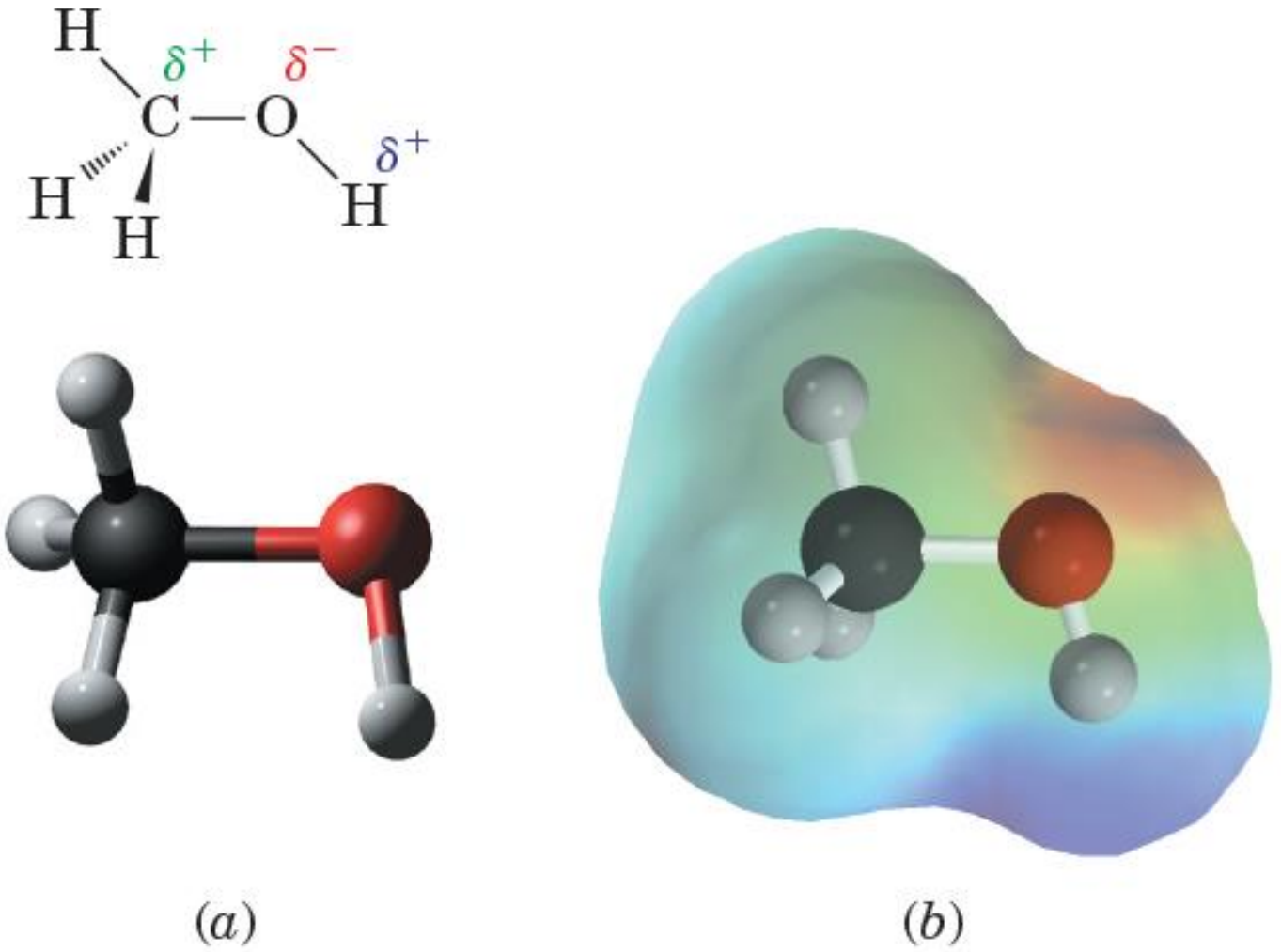
d. 1,2,3-ethanediol

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Physical Properties

Figure 13.2 Alcohols are polar molecules.

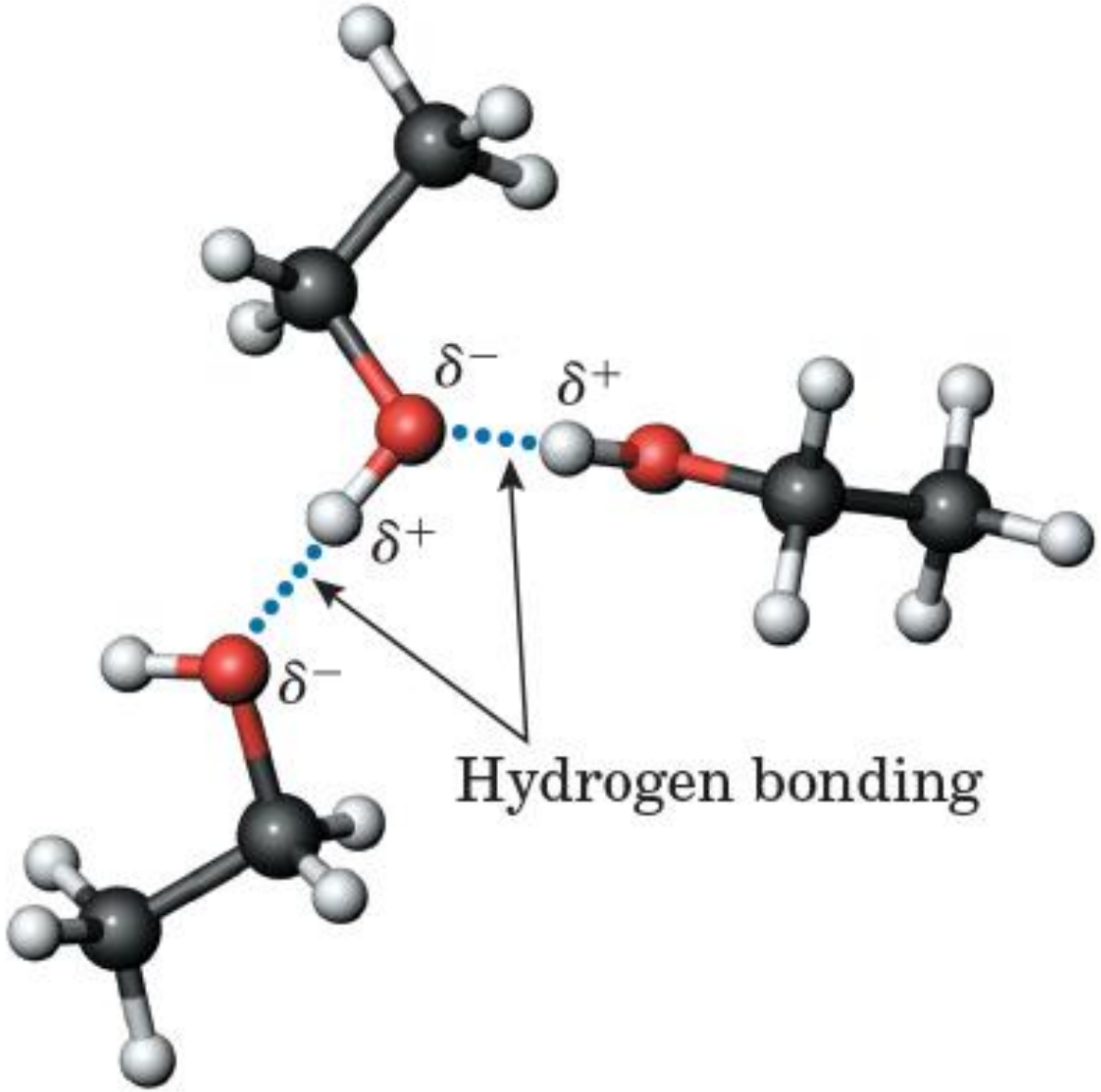
- The C–O and O–H bonds are both polar covalent.



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Figure 13.3

In the liquid state, alcohols associate by hydrogen bonding.



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Table 13.1 Boiling Points and Solubility in Water of Sets of Alcohols, and Alkanes of Similar Molecular Weight

Structural Formula	Name	Molecular Weight (amu)	Boiling Point (°C)	Solubility in Water
CH_3OH	methanol	32	65	Unlimited
CH_3CH_3	ethane	30	89	Insoluble
CH_3CH_2OH	ethanol	46	78	Infinite
$CH_3CH_2CH_3$	propane	44	42	Insoluble
$CH_3CH_2CH_2OH$	1-propanol	60	97	Infinite
$CH_3CH_2CH_2CH_3$	butane	58	0	Insoluble
$CH_3CH_2CH_2CH_2OH$	1-butanol	74	117	8 g/100 g
$CH_3CH_2CH_2CH_2CH_3$	pentane	72	36	insoluble

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EXERCISE

When compared to alkanes, the boiling point of alcohols is _____ because they undergo _____ bonding.

a. the same; covalent

b. less; dipole-dipole

c. higher; hydrogen

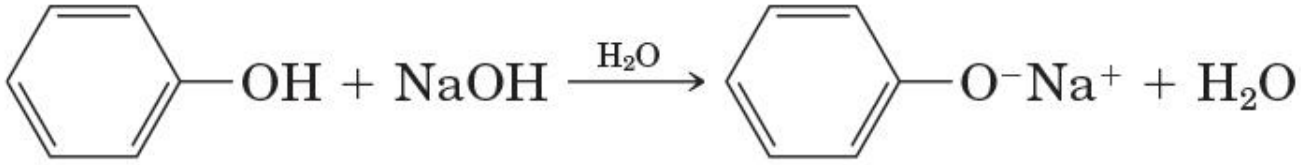
d. more; nonpolar

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Acidity of Alcohols

Alcohols have about the same pK_a values as water.

- ❑ Aqueous solutions of alcohols have the same pH as that of pure water.
- ❑ Alcohols and phenols both contain an –OH group.
- ❑ Phenols are weak acids and react with NaOH and other strong bases to form water-soluble salts.



Phenol

Sodium phenoxide
(a water-soluble salt)

- Alcohols are weaker acids than phenols and do not react in this manner.

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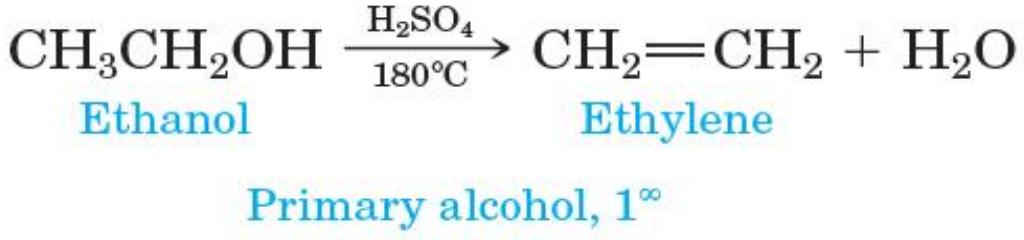
Dehydration

Dehydration: Elimination of a molecule of water from adjacent carbon atoms gives an alkene.

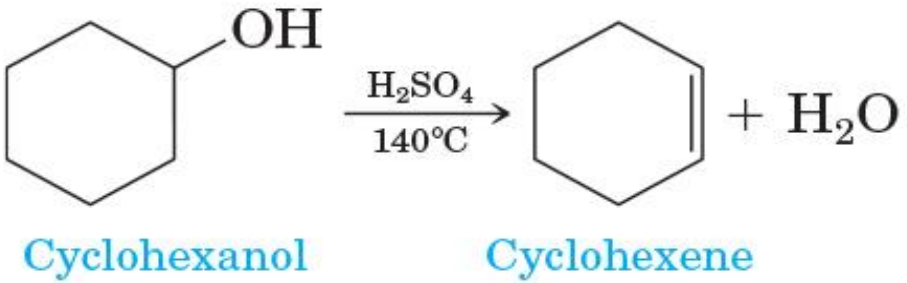
- Dehydration is most often brought about by heating an alcohol with either 85% H_3PO_4 or concentrated H_2SO_4.
  - 1° alcohols are the most difficult to dehydrate and require temperatures as high as 180° C.
  - 2° alcohols undergo acid-catalyzed dehydration at somewhat lower temperatures.
  - 3° alcohols generally undergo acid-catalyzed dehydration at temperatures only slightly above room temperature.

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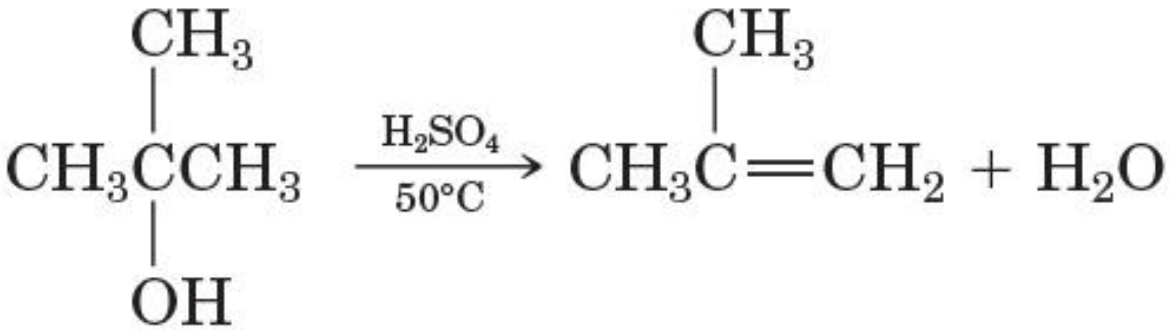
Dehydration



Primary alcohol, 1°



Secondary alcohol, 2°

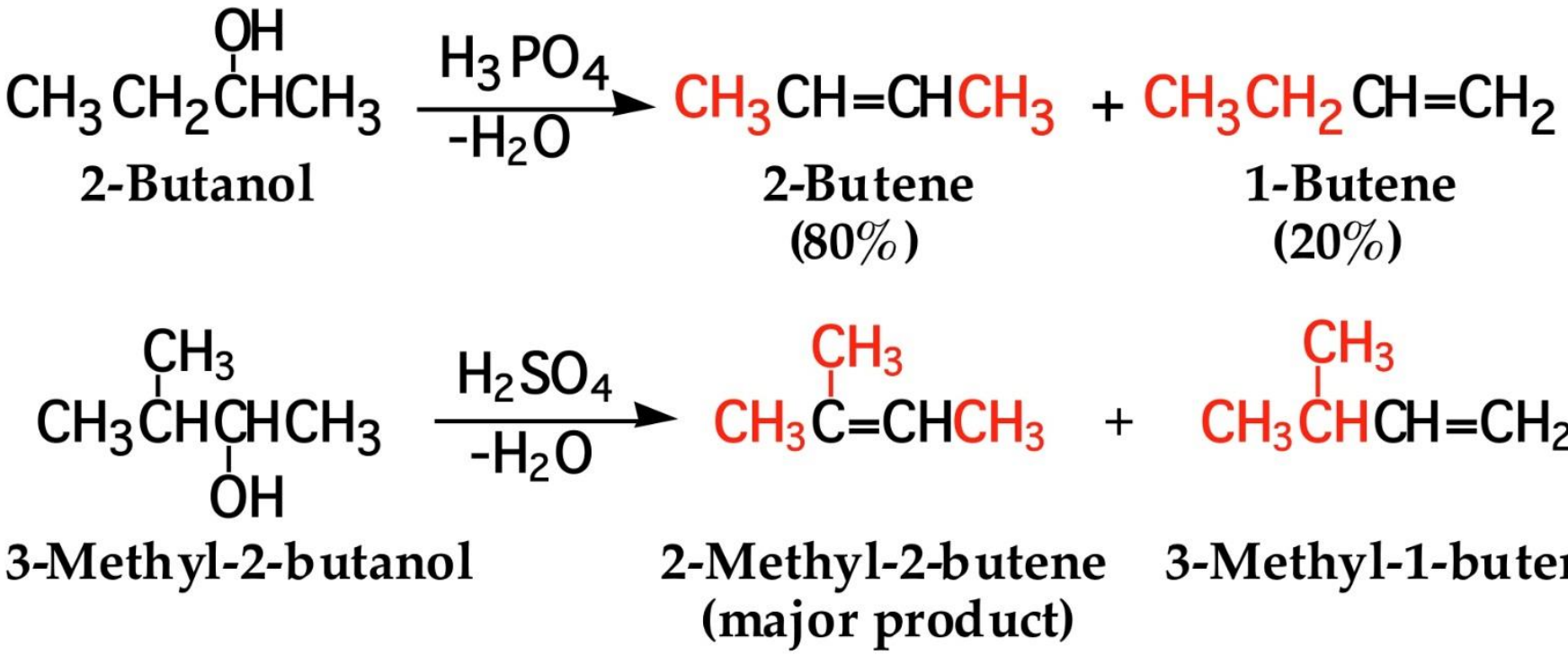


Tertiary alcohol, 3°

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When isomeric alkenes are obtained, the alkene having the greater number of alkyl groups on the double bond generally predominates.

- Examples:

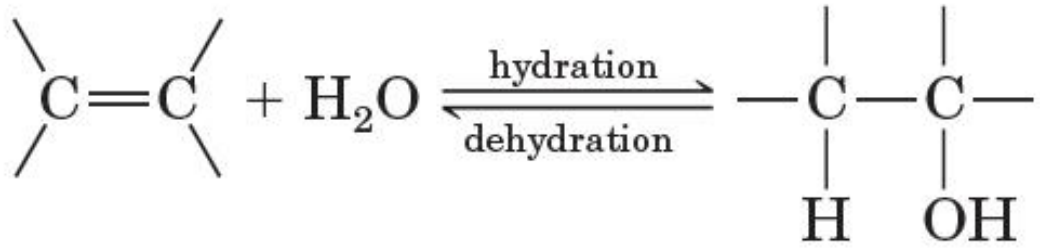


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Dehydration-Hydration

Acid-catalyzed hydration of alkenes to give alcohols and acid-catalyzed dehydration of alcohols to give alkenes are competing reactions.

- The following acid-catalyzed equilibrium exists.



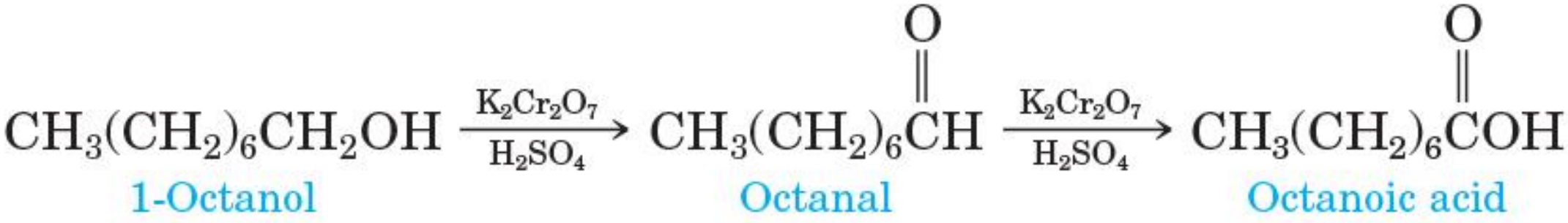
  - In accordance with Le Chatelier's principle, large amounts of water favor alcohol formation, whereas removal of water from the equilibrium mixture favors alkene formation.

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Oxidation

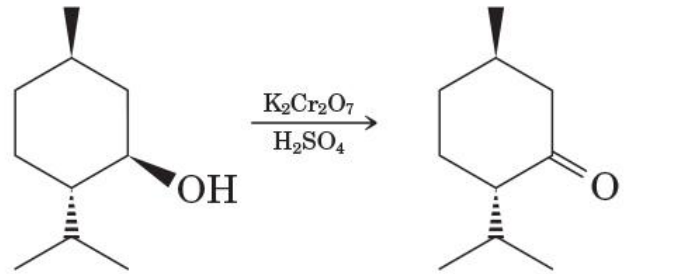
Oxidation of a 1° alcohol gives an aldehyde or a carboxylic acid, depending on the experimental conditions.

- Oxidation of a 1° alcohol to a carboxylic acid is commonly carried out using potassium dichromate, $K_2Cr_2O_7$, in aqueous sulfuric acid.



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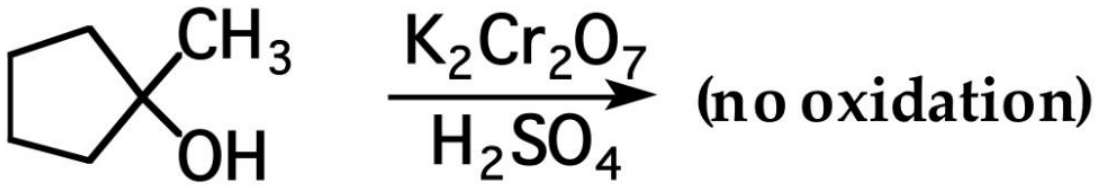
- Oxidation of a 2° alcohol gives a ketone.



2-Isopropyl-5-methyl-cyclohexanol (Menthol)

2-Isopropyl-5-methyl-cyclohexanone (Menthone)

- Tertiary alcohols are resistant to oxidation. In the presence of an acid catalyst they are prone to dehydration.



1-Methyl-cyclopentanol

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EXERCISE

One of the most important properties of phenols is their:

a. acidity.

b. basicity.

c. reactivity.

d. combustion.

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EXERCISE

Which of the following methods is used to produce alcohols?

a. Hydration of alkenes

b. Hydration of alkanes

c. Addition of —OH group by reacting an alkane with NaOH

d. Oxidation of aldehydes

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EXERCISE

What does the oxidation of a primary alcohol with a limited amount of a mild oxidizing agent produce?

a. Carboxylic acid

b. Aldehyde

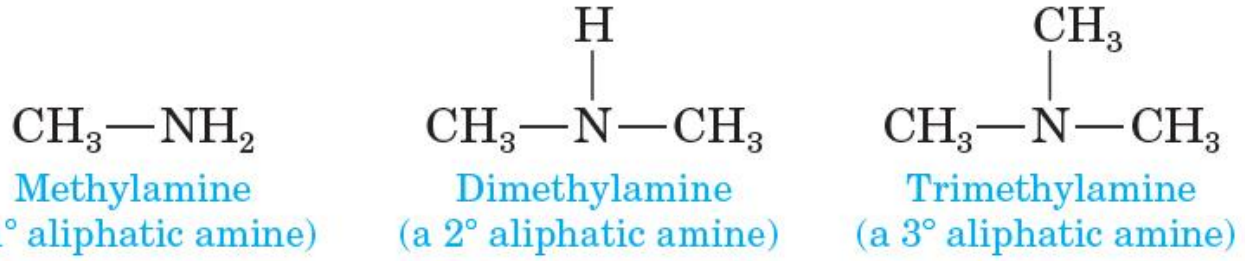
c. Ketone

d. Ether

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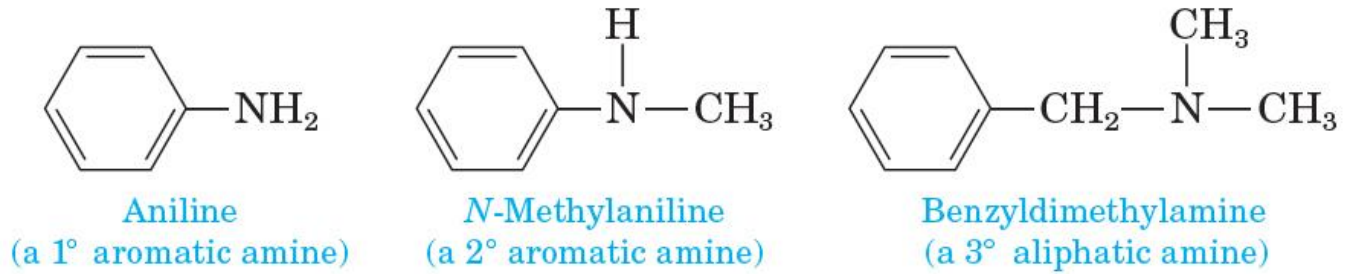
AMINES

- Structure and Classification: Amines are classified as 1° , 2° , or 3° depending on the number of carbon groups bonded to nitrogen.



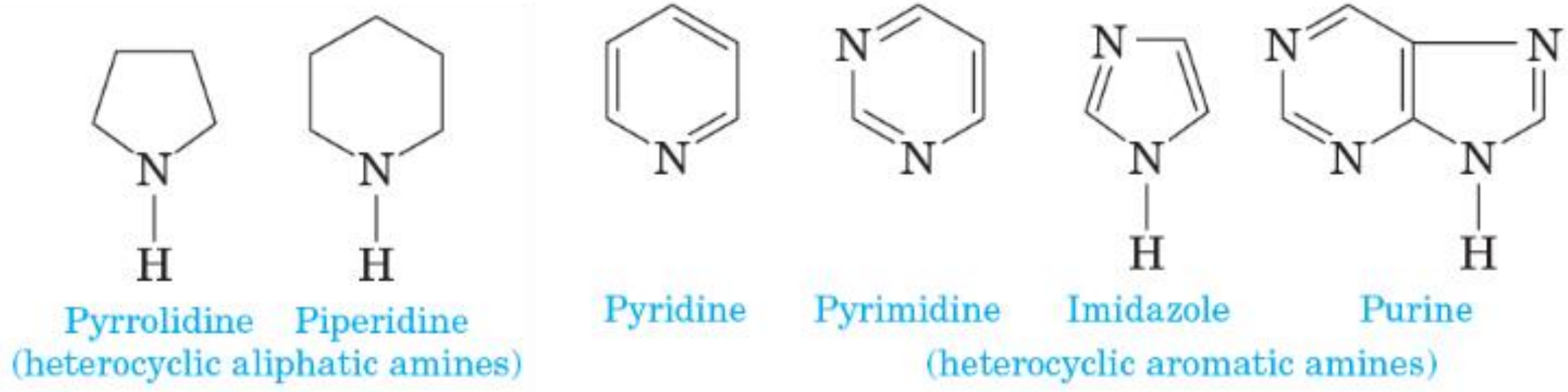
Aliphatic amine: All carbons bonded to nitrogen are derived from alkyl groups. See the three above.

Aromatic amine: One or more of the groups bonded to nitrogen are aryl groups.



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- **Heterocyclic amine**: An amine in which the nitrogen atom is part of a ring.
- **Heterocyclic aliphatic amine:** A heterocyclic amine in which the ring is saturated (has no C=C bonds).
- **Heterocyclic aromatic amine:** The amine nitrogen is part of an aromatic ring.

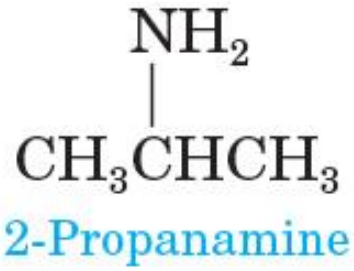


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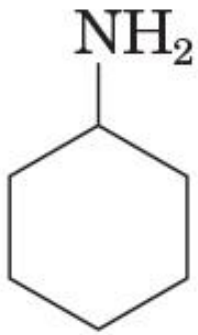
Nomenclature

IUPAC names

- ❑ We derive IUPAC names for aliphatic amines just as we did for alcohols.
- ❑ Drop the final **-e** of the parent alkane and replace it by **-amine.**
- ❑ Use a number to locate the amino group on the parent chain.



2-Propanamine



Cyclohexanamine

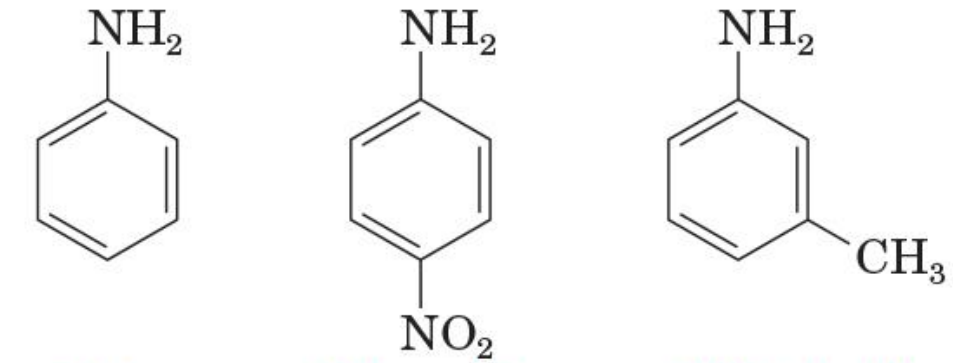


1,6-Hexanediamine

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IUPAC names (cont'd)

- IUPAC nomenclature retains the common name aniline for $C_6H_5NH_2$, the simplest aromatic amine.
- Name simple derivatives of aniline by using numbers to locate substituents or, alternatively, use the prefixes *ortho* (*o*), *meta* (*m*), and *para* (*p*).
- Several derivatives of aniline have common names that are still widely used; among them is toluidine:



Aniline ($C_6H_5NH_2$) | 4-Nitroaniline (*p*-Nitroaniline) | 3-Methylaniline (*m*-Toluidine)

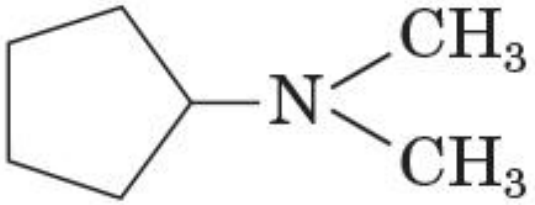
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IUPAC names (cont'd)

- Name unsymmetrical secondary and tertiary amines as *N*-substituted primary amines.
- Take the largest group bonded to nitrogen as the parent amine.
- Name the smaller group(s) bonded to nitrogen, and show their location on nitrogen by using the prefix *N*- (indicating that they are bonded to nitrogen).



N-Methylaniline

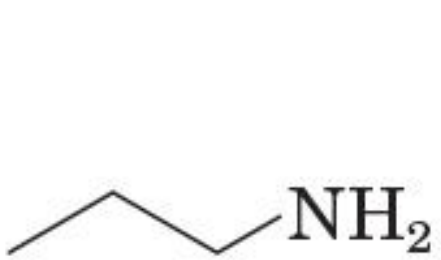


N,*N*-Dimethyl-cyclopentanamine

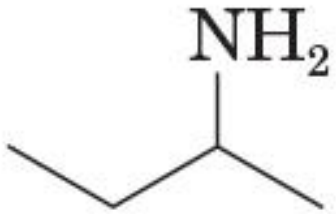
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Common names

- For most aliphatic amines, list the groups bonded **to** nitrogen in alphabetical order in one word ending in the suffix **-amine**.



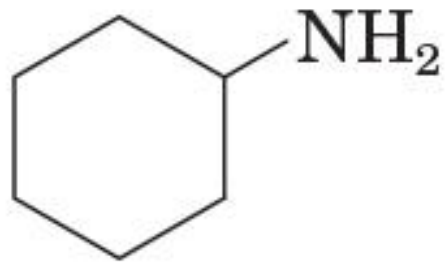
Propylamine



sec-Butylamine



Diethylmethylamine



Cyclohexylamine

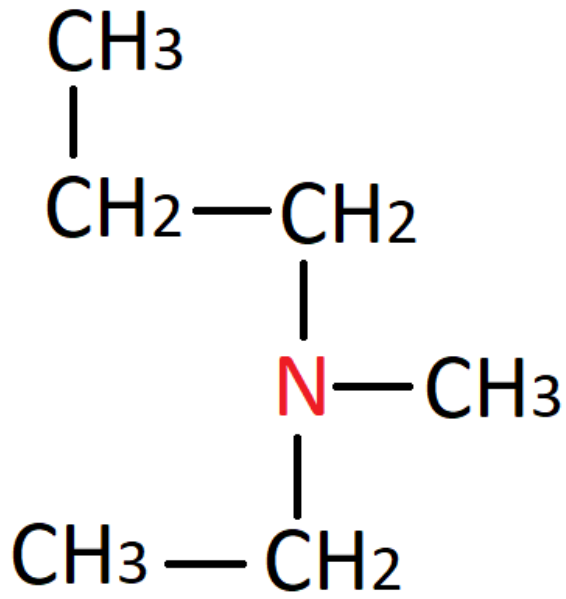
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EXERCISE

Write the IUPAC name of each organic compound.

1. $CH_3(CH_2)_5NH_2$

1-Hexanamine

2. 

N-ethyl-N-methyl-1-propanamine

Physical Properties

Like ammonia, low-molecular-weight amines have very sharp, penetrating odors.

- Trimethylamine, for example, is the pungent principle in the smell of rotting fish.
- Two other particularly pungent amines are 1,4-butanediamine (putrescine) and 1,5-pentanediamine (cadaverine).



1,4-Butanediamine (Putrescine)

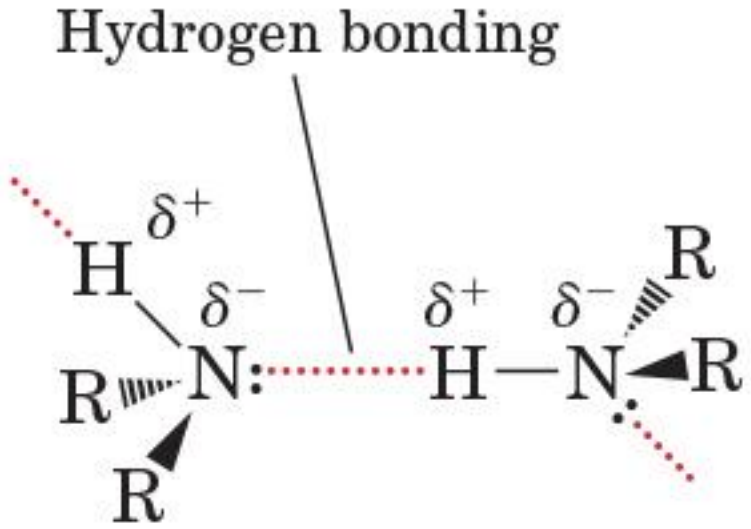


1,5-Pentanediamine (Cadaverine)

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Figure 15.1 Amines are polar compounds:

- Both 1° and 2° amines have N–H bonds, and can form hydrogen bonds with one another.
- 3° amines have no N–H bond and cannot form hydrogen bonds with one another.



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Physical Properties

- An N–H---N hydrogen bond is weaker than an O–H---O hydrogen bond, because the difference in electronegativity between N and H (3.0 – 2.1 = 0.9) is less than that between O and H (3.5 – 2.1 = 1.4).
- We see the effect of hydrogen bonding between molecules of comparable molecular weight by comparing the boiling points of ethane, methanamine, and methanol.

	CH_3CH_3	CH_3NH_2	CH_3OH
Molecular weight (amu)	30.1	31.1	32.0
Boiling point (°C)	-88.6	-6.3	65.0

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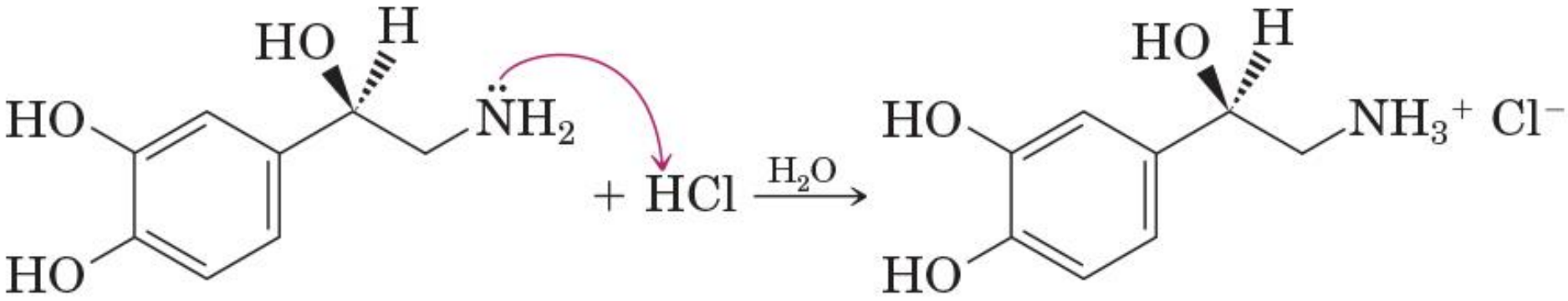
- All classes of amines form hydrogen bonds with water and are more soluble in water than are hydrocarbons of comparable molecular weight.
- Most low-molecular-weight amines are completely soluble in water.
- Higher-molecular-weight amines are only moderately soluble in water or are insoluble.

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Reactions of Amines

The most important chemical property of amines is their basicity.

- Amines, whether soluble or insoluble in water, react quantitatively with strong acids to form water-soluble salts.



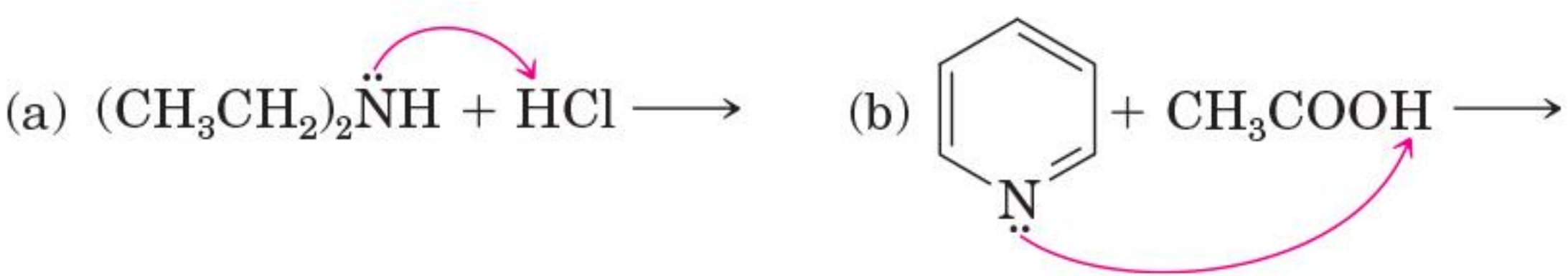
(*R*)-Norepinephrine
(only slightly soluble in water)

(*R*)-Norepinephrine hydrochloride
(a water-soluble salt)

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EXERCISE

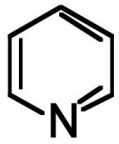
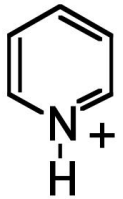
Complete each acid–base reaction and name the salt formed.



ANSWER

(a) $(CH_3CH_2)_2NH + HCl \longrightarrow (CH_3CH_2)_2NH_2^+Cl^-$

Diethylammonium chloride

(b)  $+ CH_3COOH \longrightarrow$  CH_3COO^-

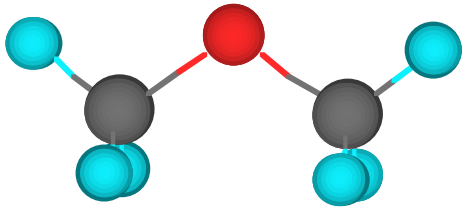
Pyridinium acetate

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Ethers

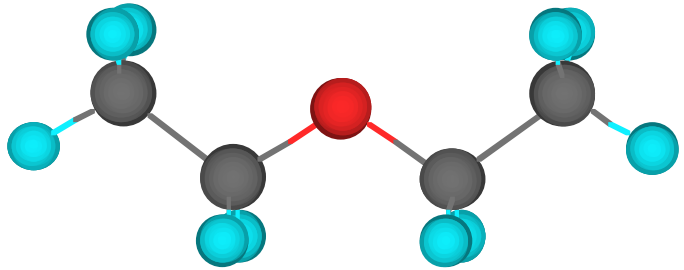
The functional group of an **ether** is an oxygen atom bonded to two carbon atoms.

- The simplest ether is dimethyl ether.
- The most common ether is diethyl ether.



CH_3-O-CH_3

Dimethyl ether



CH_3CH_2-O-CH_2CH_3

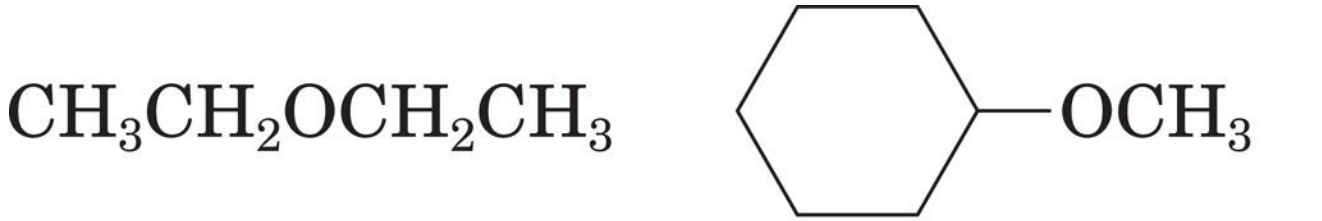
Diethyl ether

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Nomenclature

Although ethers can be named according to the IUPAC system, chemists almost invariably use common names for low-molecular-weight ethers.

- Common names are derived by listing the alkyl groups bonded to oxygen in alphabetical order and adding the word "ether."
- Alternatively, name one of the groups on oxygen as an alkoxy group.



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Cyclic ether: An ether in which one of the atoms in a ring is oxygen.

- Cyclic ethers are also known by their common names.
- Ethylene oxide is an important building block for the organic chemical industry. It is also used as a fumigant in foodstuffs and textiles, and in hospitals to sterilize surgical instruments.
- Tetrahydrofuran is a useful laboratory and industrial solvent.



Ethylene oxide



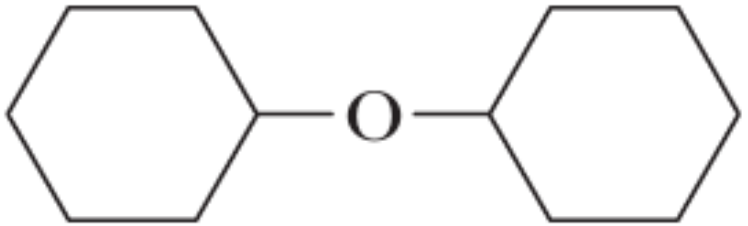
Tetrahydrofuran
(THF)

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EXERCISE

Write the IUPAC name of each organic compound.

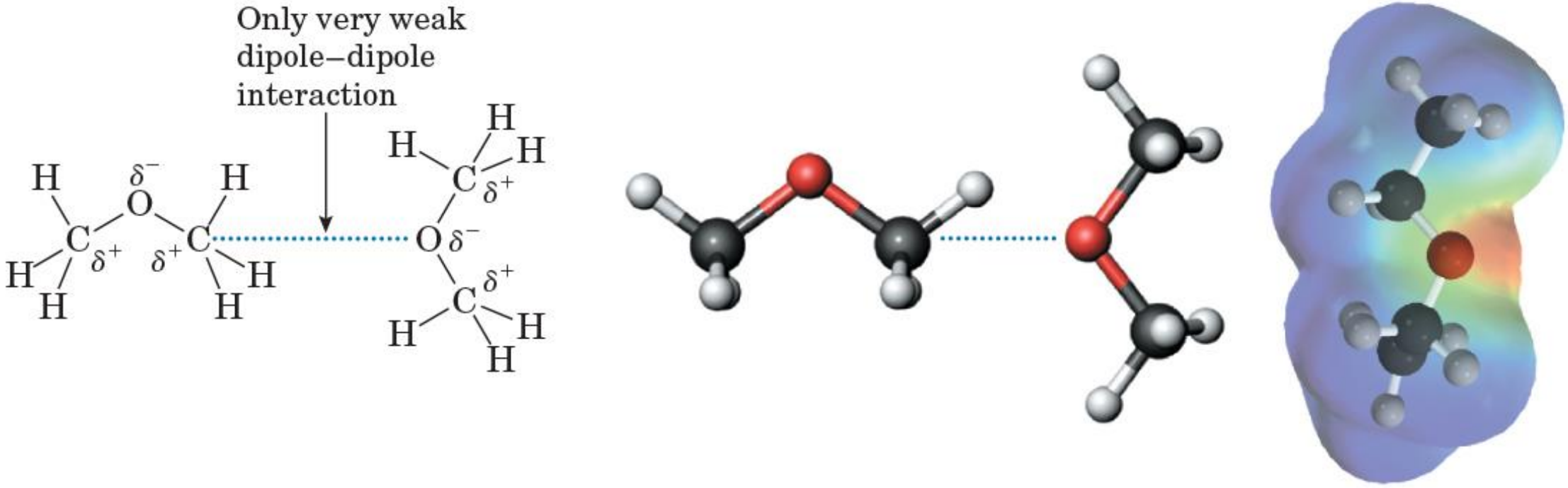
1. $CH_3CH_2CH_2CH_2CH_2$-O-CH_3 1-methoxypentane

2. 

cyclohexoxycyclohexane

Physical Properties

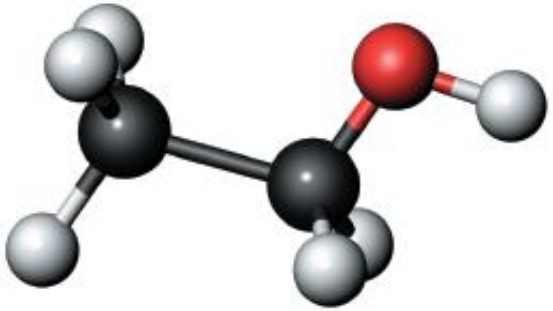
Figure 13.5 Ethers are polar molecules in which oxygen bears a partial negative charge and each carbon bonded to it bears a partial positive charge.



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However, only weak forces of attraction exist between ether molecules in the pure liquid.

- Consequently, boiling points of ethers are close to those of hydrocarbons of similar molecular weight.
- Ethers have lower boiling points than alcohols of the same molecular formula.



CH_3CH_2OH

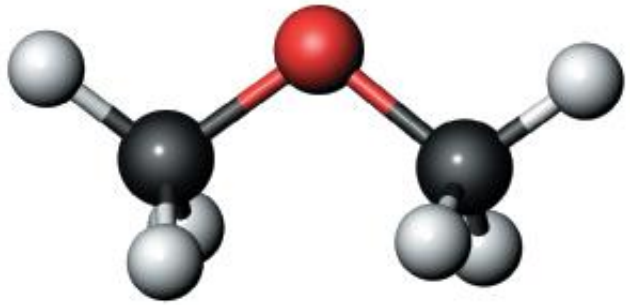
Ethanol

bp 78°C

CH_3OCH_3

Dimethyl ether

bp –24°C

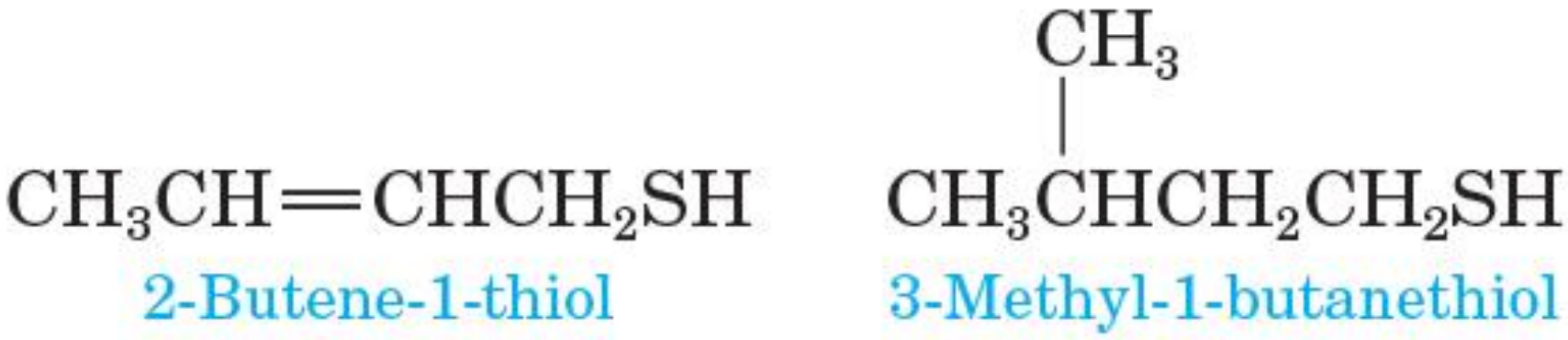


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Thiols

Thiol: A compound containing an **-SH (sulfhydryl group).**

- The most outstanding property of low-molecular-weight thiols is their stench.
- They are responsible for smells such as those from rotten eggs and sewage.
- The scent of skunks is due primarily to these two thiols.



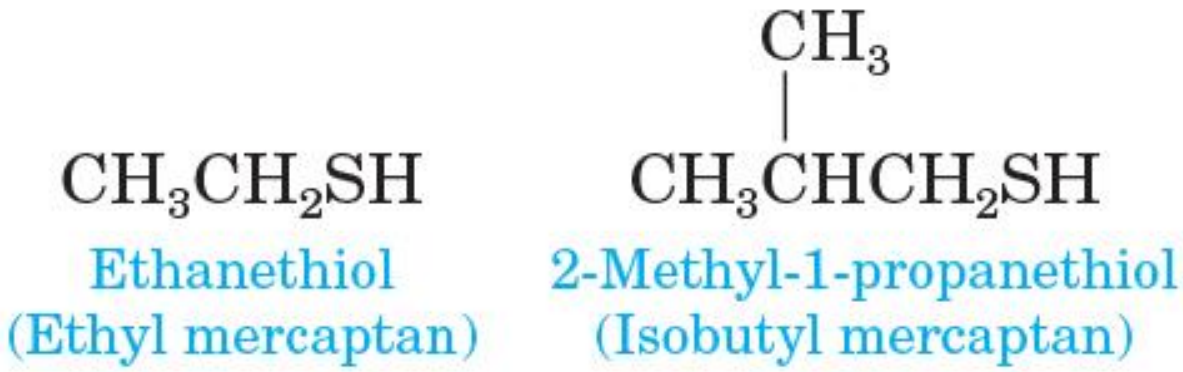
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Nomenclature

IUPAC names are derived in the same manner as are the names of alcohols.

- To show that the compound is a thiol, the final **-e** of the parent alkane is retained and the suffix **-thiol** added.

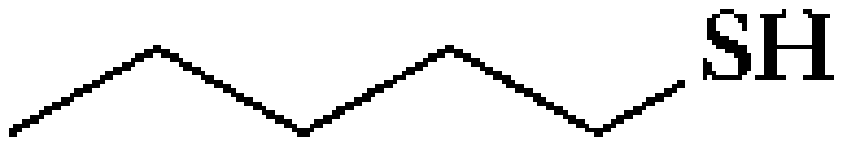
Common names for simple thiols are derived by naming the alkyl group bonded to –SH and adding the word "**mercaptan**."



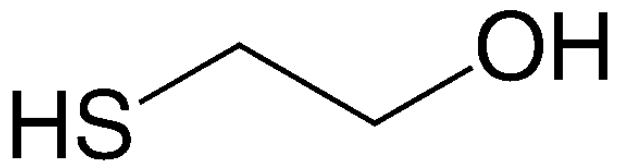
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EXERCISE

Write the IUPAC name of each organic compound.

1. 

1-pentanethiol

2. 

2-mercaptoethanol

Physical Properties

Because of the small difference in electronegativity between sulfur and hydrogen (2.5 – 2.1 = 0.4), an S–H bond is nonpolar covalent.

- Thiols show little association by hydrogen bonding.
- Thiols have lower boiling points and are less soluble in water and other polar solvents than alcohols of similar molecular weight.

TABLE 13.2 Boiling Points of Three Sets of Thiols and Alcohols with the Same Number of Carbon Atoms

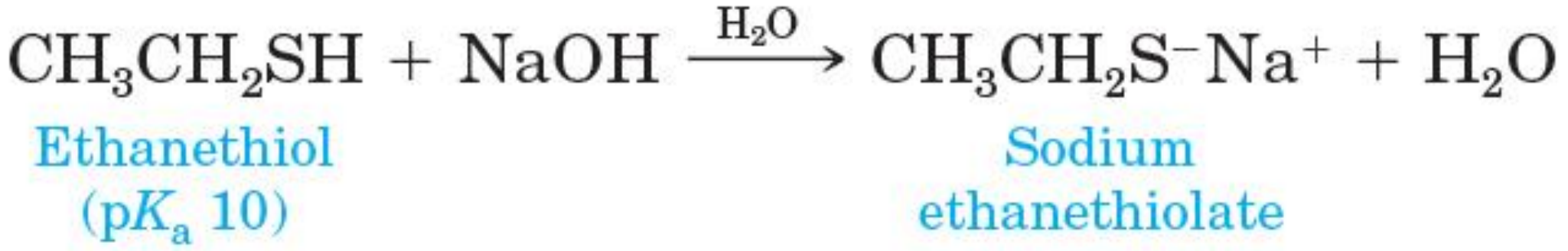
Thiol	Boiling Point (°C)	Alcohol	Boiling Point (°C)
methanethiol	6	methanol	65
ethanethiol	35	ethanol	78
1-butanethiol	98	1-butanol	117

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Acidity of Thiols

Thiols are weak acids (pK_a 10) and are comparable in strength to phenols (pK_a 10).

- Thiols react with strong bases such as NaOH to form water-soluble thiolate salts.

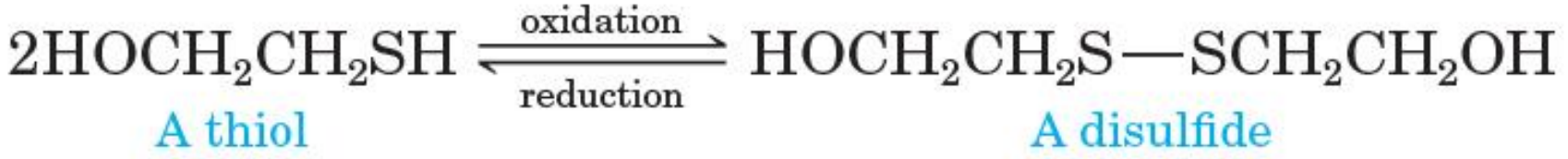


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Oxidation of Thiols

The most common reaction of thiols in biological systems is their oxidation to disulfides, the functional group of which is a **disulfide (–S–S–) bond.**

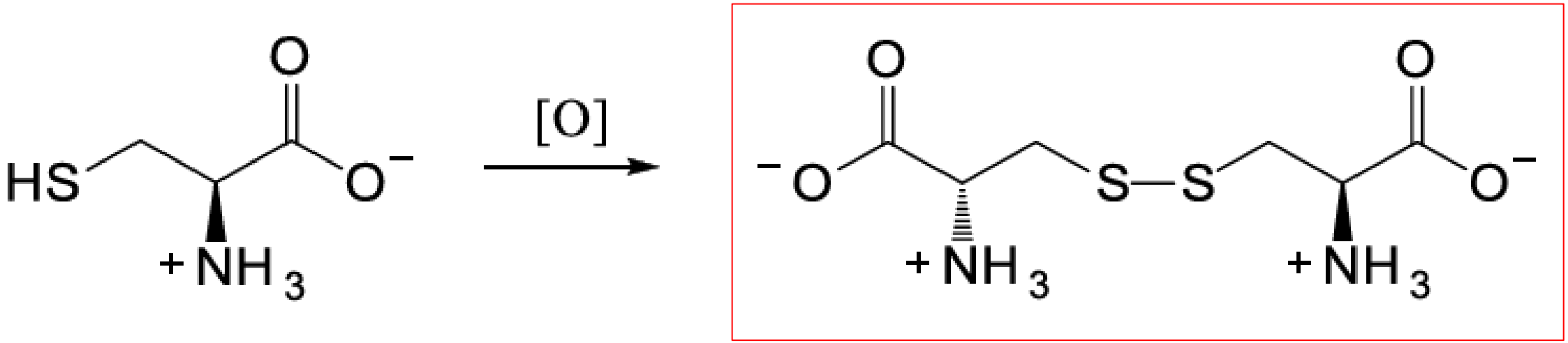
- Thiols are readily oxidized to disulfides by O_2.
- They are so susceptible to oxidation that they must be protected from contact with air during storage.
- Disulfides, in turn, are easily reduced to thiols by several reducing agents including H_2 in the presence of a transition metal catalyst.



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EXERCISE

Cysteine is a functional building block of keratin, which is found in hair and nails. Disulfide bonds can form between the cysteine segments in keratin. Show the product formed from a reaction of this type.



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